

Reviving Online Sales: Addressing Order Decline in a Biryani Cloud Kitchen

A Mid-Term Report

Name: Sharmila L

Roll number: 21f2xxxxxx



*IITM BS Degree Program,
Indian Institute of Technology, Madras,
Chennai Tamil Nadu, India, 600036*

Table of contents

Reviving Online Sales: Addressing Order Decline in a Biryani Cloud Kitchen.....	1
Table of contents.....	2
1. Executive Summary.....	3
2. Proof of originality of the Data.....	3
3. Metadata-.....	3
3.1 Detailed Sales Dataset:.....	4
3.3 Competitor Dataset :.....	4
4. Descriptive Statistics.....	5
4.1 Local Demand and Delivery Distance.....	5
4.2 Customer Ordering Behavior.....	6
4.3 Menu-Level Insights.....	6
4.4 Competitor Landscape.....	6
5. Detailed Explanation of Analysis Process/Method-.....	6
5.1 Data Cleaning and Preprocessing.....	6
5.2 Analysis Process/Method.....	7
5.2.1 Time Series Analysis.....	8
5.2.2 Distribution Analysis and Transformation.....	8
5.2.3 Menu Analysis and Co-Occurrence Network.....	8
5.2.4 Text Analytics and Sentiment Classification.....	8
5.2.5 Customer Retention and Segmentation.....	9
6. Results and Findings.....	9
6.1 Time Series and Temporal Patterns.....	9
6.2 Menu and Sentiment Analysis.....	11
6.3 Customer Behavior Analysis.....	12

1. Executive Summary

This project is undertaken in collaboration with Ax-Wxxx Bxxxxxxx, a cloud kitchen located in Othakadai, Madurai. As a B2C business, it primarily relies on Zomato for online sales and operates mainly during lunch hours. Despite offering competitive pricing in the biryani segment, the business has seen limited sales growth since mid-2023, with a flat or declining trend in monthly online orders. Additional challenges include low brand recognition, the absence of a structured customer retention strategy, and operational constraints during peak hours.

Specifically, Zomato's algorithm has occasionally applied discount offers without the restaurant's consent, affecting profitability. Furthermore, delivery partner shortages during lunch rushes have led to order delays or cancellations, with up to three orders per week reported as unassigned.

To address these issues, the following key datasets were analyzed: sales data from the Zomato merchant dashboard, customer data inferred using encoded Customer IDs, and competitor data collected via web scraping from Google Maps and Zomato search results. The sales dataset was structured and required basic cleaning, while the scraped competitor data initially included 242 entries, which were filtered down to 41 direct competitors based on proximity, cuisine type, and delivery feasibility. Exploratory Data Analysis (EDA) was conducted to understand both numerical and categorical variables across all datasets.

The analysis methodology involved descriptive statistics, time-series modeling using SARIMA (to capture weekly sales seasonality), and customer behavior analysis through churn rate, bill value distribution, and sentiment classification. Preliminary results revealed higher average order values on weekends despite greater volume on weekdays, along with a high customer churn rate (96%). Competitor insights were derived based on pricing relative to Ax-Wxxx's hero item, delivery times, and menu overlap. These insights are expected to guide strategic interventions to improve Ax-Wxxx Bxxxxxxx's customer retention, brand positioning, and overall sales performance in the final phase of the project.

2. Proof of originality of the Data

—Intentionally Left Blank—

3. Metadata-

The following datasets were collected and used to address key business objectives:

3.1 Detailed Sales Dataset:

This provides an itemized breakdown of each order, including discounts, ratings, and detailed reviews, covering the period from September 2024 to February 2025. It consists of 1,081 rows and 29 features, along with an additional sheet containing lifetime sales records from 27 April 2023 to 31 January 2025. It is critical for understanding the customer base by analyzing order patterns, identifying factors influencing repeat purchases, and developing targeted solutions.

This dataset directly supports the investigation of **low customer retention, order behavior, and price sensitivity**, which are central to the business challenge. It includes the following columns:

- Restaurant ID – Unique ID for the restaurant (Common in Sheet 1 & 2)
- Restaurant name – Name of the restaurant
- Subzone – Local area where the restaurant is located
- City – City of the restaurant
- Order ID – Unique ID for each order (Common in Sheet 1 & 2)
- Order Placed At – Date and time when the order was placed (Common in Sheet 1 & 2)
- Order Status – Final outcome of the order (e.g., Delivered, Cancelled)
- Delivery – Mode of delivery (e.g., Zomato Delivery)
- Distance – Distance between restaurant and customer
- Items in order – Food items included in the order
- Instructions – Special notes or requests from the customer
- Discount construct – Type of discount applied (e.g., Buy 1 Get 1)
- Bill subtotal – Total cost before any discounts
- Packaging charges – Additional cost for food packaging
- Restaurant discount (Promo) – Promotional discount from the restaurant
- Restaurant discount (Flat offs, Freebies & others) – Other discounts like flat off
- Gold discount – Discount from Zomato Gold membership
- Brand pack discount – Discount applied on combo/brand offers
- Total – Final bill amount after all discounts (Common in Sheet 1 & 2)
- Rating – Customer's star rating (e.g., out of 5)
- Review – Text review from the customer
- Cancellation / Rejection reason – Reason for order cancellation or rejection
- Restaurant compensation (Cancellation) – Amount paid back for cancelled order
- Restaurant penalty (Rejection) – Fine charged to the restaurant for rejecting an order
- KPT duration (minutes) – Time taken to prepare the order in the kitchen
- Rider wait time (minutes) – Time delivery person waited at the restaurant
- Order Ready Marked – Whether the order was marked ready properly
- Customer complaint tag – Type of complaint raised by customer
- Customer ID – Unique identifier for the customer

3.3 Competitor Dataset :

No structured competitor data was readily available. Therefore, restaurant listings were scraped from Google Maps and Zomato search results for the Othakadai region using Beautiful Soup Python Library. These were merged to study the dominant biryani types, restaurant specialization, and direct competitors' performance. The final dataset contains 13 features across 242 restaurants. It informs Ax-Wxxx's **positioning strategy** by revealing how it compares to nearby competitors in terms of **pricing, delivery speed, and menu offerings**. It includes the following columns:

- restaurant_name – Name of the restaurant

- rating – Average customer rating (e.g., out of 5)
- Restaurant Type –Veg or Non Veg restaurant or Dine-in or Fast food vendor
- cuisine_types – Types of cuisines offered (e.g., Biryani, Chinese)
- price_for_one – Approximate cost for one person
- delivery_time – Estimated delivery time in minutes
- address – Physical address of the restaurant
- lat/long – Latitude and longitude coordinates of the restaurant location
- bestseller – Most popular or frequently ordered item(s)
- bir_type – Type of biryani offered (e.g., Chicken, Mutton, Egg)
- price – Price of the listed item
- review_count – Total number of customer reviews received
- Comments - Special offline information obtained through interview

4. Descriptive Statistics

Out of 1,081 total orders, 1,028 were successfully delivered. The **average order value** stands at approximately ₹242.60, while the **median is ₹170**, suggesting a right-skewed distribution driven by a few high-value outliers (maximum: ₹1,450). This indicates that while most customers order conservatively, occasional large orders significantly raise the average.

In terms of Kitchen Preparation Time (**KPT**), operations appear efficient:

- **75% of orders** were prepared in under **8.7 minutes**,
- **Median KPT** is just **5.8 minutes**,
- Maximum KPT is 30.7 minutes—indicating that delays are rare.

The **rider wait time** shows a median of **2.2 minutes**, reflecting good coordination between kitchen readiness and delivery pickups.

4.1 Local Demand and Delivery Distance

The majority of orders came from within a **4 km radius**, with an even spread across 1–5 km. This shows strong hyperlocal demand centered around the restaurant's location. The most frequent delivery distance was exactly 4 km (164 orders).

4.2 Customer Ordering Behavior

The most commonly ordered item was **2 x Chicken Biryani [Regular]** (226 times). From September 2024 to February 2025, 1,678 biryanis were delivered in total. **Starters** followed as the next most preferred category (208 orders), with **Chicken 65 (500g)** being the least ordered among its variants. Still, **regular Chicken 65** remains a popular side, even when it appears in combos—suggesting a strong standalone preference.

Customer retention analysis reveals that **76.6%** are one-time buyers, with only **23.4%** repeat customers. Notably, one loyal customer placed **20 orders** over six months—indicating a potential segment of high-LTV individuals if properly nurtured.

4.3 Menu-Level Insights

While the budget-friendly **Hero Biryani** drives volume, higher-rated items like the *Egg Biryani Combo* and *Mini Bucket Biryani* show strong customer satisfaction and upsell potential.

The most frequently used discount was *Buy 1 Get 1 – ₹149* (291 orders), which likely encouraged bulk purchases and influenced order value patterns.

4.4 Competitor Landscape

Most competitors price their biryanis **above ₹200**, making Ax-Wxxx's lower-priced offerings more attractive to price-sensitive customers. The **<₹200 segment** is less competitive, offering strategic opportunity. Among competitors, **New Mass Hotel** stands out with a similar pricing model and a solid **4.2 rating**, positioning it as a direct threat.

These descriptive statistics provide a foundational understanding of **customer preferences, operational efficiency, and market positioning**. By analyzing order values, frequency, delivery distance, and product popularity, we can better tailor the menu, plan offers, and optimize kitchen performance. This insight is also crucial for identifying high-value customer segments and capturing untapped local demand — addressing the core challenges of **low retention, limited brand recognition, and stagnant sales growth**.

5. Detailed Explanation of Analysis Process/Method-

5.1 Data Cleaning and Preprocessing

The initial stage of analysis involved comprehensive data cleaning to ensure the reliability and analytical usability of the Zomato sales dataset. This process included the removal of null values (NaN), ambiguous symbols (e.g., '?'), and irrelevant or redundant attributes such as Restaurant ID, Restaurant Name, Subzone, City which did not yield actionable insights. Additionally, relevant columns like *'Order Placed At'* were converted from object types to standardized datetime formats to enable time-based trend analysis. A derived binary feature for *'Weekend vs Weekday'* was also created to support temporal behavior analysis.

A derived feature was also created from the Customer ID column to distinguish between repeat and non-repeat customers. By identifying repeated customer IDs across multiple transactions, a binary classification was generated to indicate whether an order was placed by a **returning customer** or a **first-time customer**. This transformation enabled segmentation of the customer base and allowed for further analysis with variables such as order frequency, bill value, and sentiment.

In parallel, competitor data was sourced through a custom web scraping approach, as structured datasets were not publicly available. Restaurant listings were extracted from Google Maps and Zomato search results, focusing on establishments located in the Othakadai region. The initial dataset comprised 242 entries, which were systematically filtered based on a set of criteria. First, restaurants offering non-vegetarian items were retained to align with the product scope of the business under study. From this subset, further refinement was applied by selecting restaurants that explicitly listed biryani as a primary offering. Among these, restaurants specializing in Seeraga Samba or Hyderabadi biryani were excluded to maintain consistency in product comparison, and only those offering Basmati biryani were included.

Subsequent filters were applied to enhance operational comparability. Only those restaurants with a best-selling item priced under ₹400 were retained, representing affordability alignment with the target business. In addition, delivery time was used as a proxy for operational efficiency, and only those with an average delivery time under 40 minutes were included—based on customer behavior trends observed in the sales data. To ensure digital visibility and relevance, only restaurants appearing within the top 150 Zomato search results were considered. This multi-step trait-based filtration strategy ultimately resulted in a refined set of 41 restaurants, identified as direct competitors for the purpose of subsequent analysis.

242 listings → Non-veg → Biryani-focused → Basmati only → Price < ₹400 → Delivery < 40 min → Top 150 Zomato → 41 direct competitors

Finally, to simplify the complexity of multiple discount-related columns, a binary indicator was created to represent whether an order was placed with or without a discount. This transformation streamlined promotional impact analysis across customer segments.

5.2 Analysis Process/Method

To address the business challenges—namely customer retention, sales growth, and competitive positioning—a multi-method analytical framework was adopted. This included time series forecasting, statistical transformation, natural language processing, co-occurrence network analysis, and clustering. Each method was chosen based on its relevance to the specific dataset and the insights needed.

5.2.1 Time Series Analysis

Temporal patterns in customer orders were investigated using **Seasonal ARIMA (SARIMA)** modeling. The Order Date column was first converted from object to datetime format to support time-indexed analysis. To identify potential seasonality, **autocorrelation (ACF)** and **partial autocorrelation (PACF)** plots were generated, revealing prominent weekly cycles at lags of 7 and 14 days. Consequently, a SARIMA(p,d,q)(P,D,Q,s) model with a seasonal component of s = 7 was adopted to account for both trend and seasonality in the data. In addition, two lag-based features were engineered to support predictive modeling and uncover short-term temporal dependencies in customer ordering patterns.

5.2.2 Distribution Analysis and Transformation

The bill amount variable exhibited a **right-skewed distribution**, with infrequent but extreme values exceeding ₹1300. To correct for this skewness and stabilize variance, a **logarithmic transformation** was applied:

$$\log_bill = \log(bill_value + 1)$$

This transformation improved model reliability and enabled more robust statistical comparisons.

5.2.3 Menu Analysis and Co-Occurrence Network

The dataset included an 'Items in Order' column containing a list of comma-separated items per order, representing an unnormalized structure. A **custom text extraction pipeline** was developed using Python's string function, regular expressions, and conditional logic to parse individual items. These were then classified into five major categories: *Combo/BucketCombo*, *Biryani*, *Drinks*, *Starters*, and *Bucket/Padi*. To explore inter-item relationships, a **network graph analysis** was conducted using co-occurrence data, revealing frequently ordered item combinations and assessing whether existing menu bundles align with customer ordering behavior. This analysis supported insights into potential improvements in product bundling strategy.

5.2.4 Text Analytics and Sentiment Classification

Customer reviews were analyzed using Natural Language Processing (NLP). Reviews were first normalized by lowercasing, and cleaned of emojis, punctuation, and special characters using regular expressions. Further preprocessing included **stopword removal** using the NLTK library. To enable structured sentiment analysis, ratings were binned into three classes: **Positive (4–5)**, **Neutral (3)**, and **Negative (1–2)**. This allowed structured analysis of brand perception over time, and the relationship between satisfaction, menu items, and delivery performance.

5.2.5 Customer Retention and Segmentation

To study retention and lifetime value, a **cohort-based model** was used. Each customer was assigned a cohort based on their first purchase month. A cohort matrix was constructed to track retention trends and derive:

- Customer Churn Rate (CCR)
- Repeat Purchase Rate (RPR)

In addition, Customer Lifetime Value (CLV) was estimated by aggregating bill values across the lifetime of each customer and normalizing by their active duration [**CLV = (Total bill value over lifetime) / (Active months)**]. This metric identified high-value customers who could be targeted for loyalty programs. A purchase frequency distribution was also analyzed to understand ordering behavior across different customer types.

To understand ordering diversity and volume, **K-Means clustering** was performed using features like total item quantity and item variety. The resulting clusters revealed behavior-based segments like high-volume loyalists, promotion-driven buyers, and single-item infrequent users. These segments directly support menu optimization and targeted campaigns.

6. Results and Findings

6.1 Time Series and Temporal Patterns

Monthly box plots revealed a consistent trend in median bill values, with only occasional high-value outliers in May, July, and December—likely driven by festive periods or holidays. This suggests a steady baseline in customer spending throughout the year, without sharp month-to-month variations.

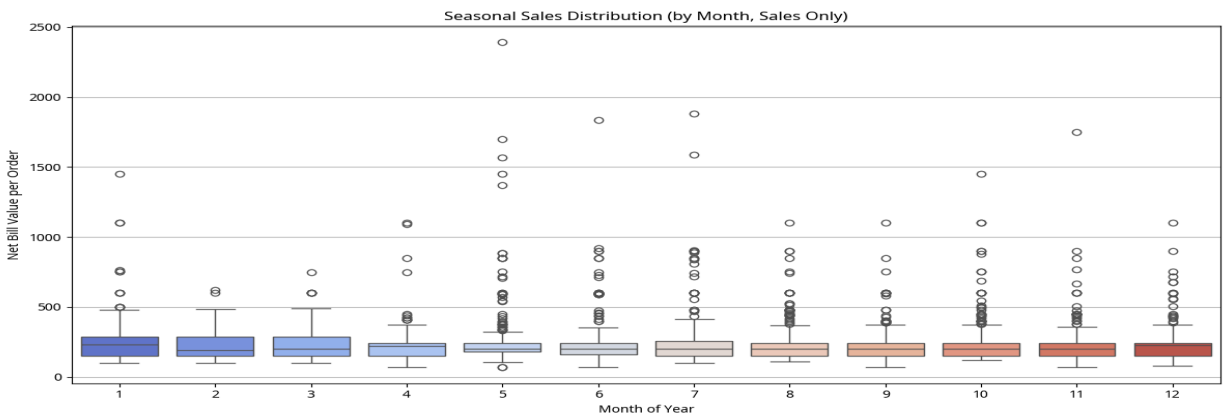


Figure 1 - Box Plot [Seasonal Sales Distribution by Month]

The SARIMA(1,0,1)(1,1,1,7) model was employed to forecast short-term demand, successfully capturing weekly seasonality. Forecasted values closely aligned with actuals in the test set, validating the presence of a strong 7-day cycle. This is particularly valuable for planning kitchen

staffing and inventory.

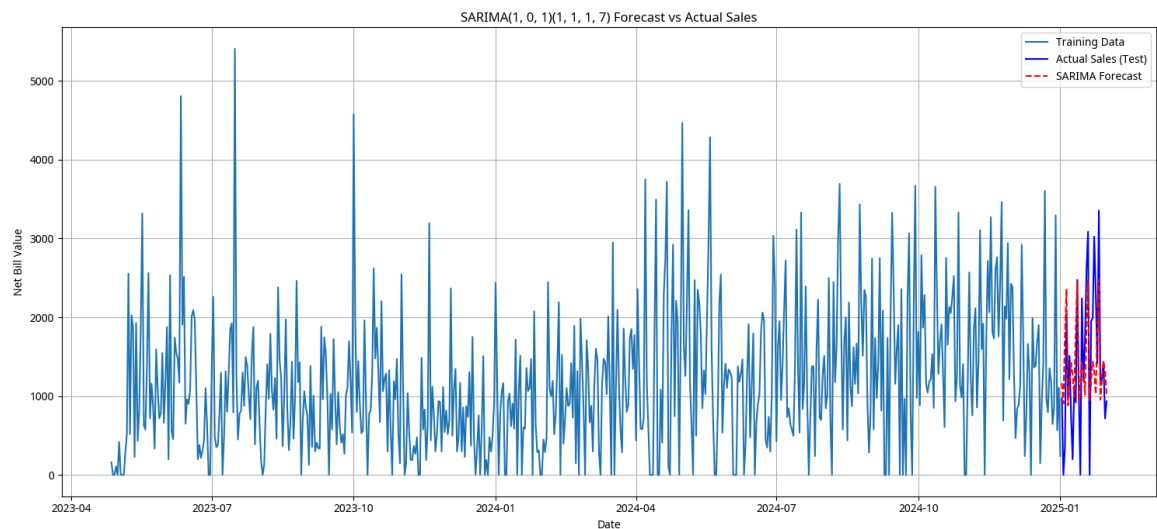


Figure 2 - Time Series Plot [SARIMA Forecast]

Further comparison of weekday vs weekend performance revealed that although **weekend orders had a higher average bill value**, the **total net sales** were significantly higher on weekdays, driven by greater order volume. This indicates distinct customer behavior: **weekdays reflect routine, high-frequency purchases**, while **weekends tend to involve higher-value, lower-frequency transactions**, likely due to family or group orders. This distinction highlights an opportunity for strategy: weekday promotions could focus on order volume, while weekend campaigns might benefit from value bundling or premium offerings.

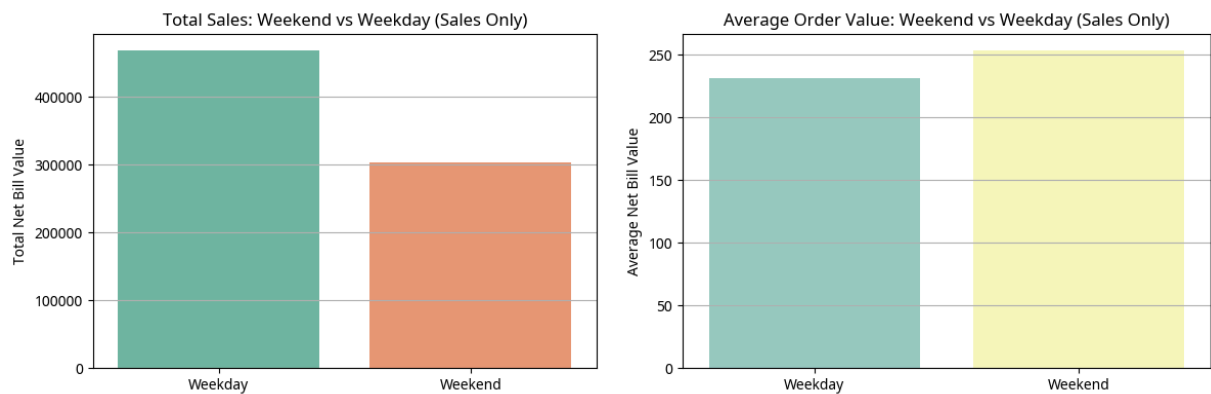


Figure 3 - Weekday Vs Weekend Plot [Left: Total Sales, Right: AOV]

6.2 Menu and Sentiment Analysis

Sales data showed that items like Special 65 Chicken Biryani (₹199) and 65 Chicken Biryani (₹189) had limited demand (175 and 8 orders respectively), despite minimal price differences.

These offerings are likely splitting customer attention without significantly boosting revenue. Similarly, low-performing items like Chicken Biryani Pack (1 Piece), Double Egg Biryani, and Chicken Biryani (Half) had minimal orders and weak or no positive sentiment. Removing these items can streamline the menu, reduce kitchen complexity, and shift focus to higher-performing dishes.

A co-occurrence heatmap revealed frequent pairing of Regular Chicken Biryani and Chicken 65—even though many combo options already include Chicken 65. This may reflect customer perceptions that combo portions are smaller, prompting separate orders. Clearly indicating portion sizes (e.g., number of pieces) instead of grams may help reduce confusion and encourage combo purchases.

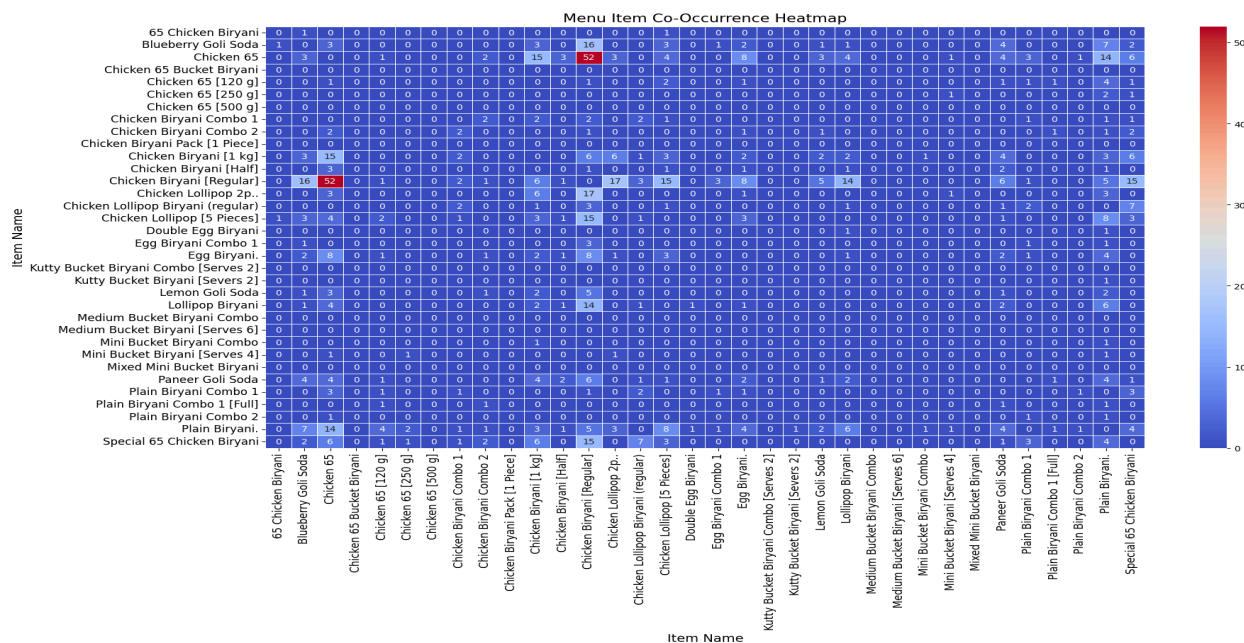


Figure 4 - Co-occurrence Heatmap [Menu Items]

However, correlation analysis showed weak or no positive relationship between these items, implying that ordering one does not predict ordering the other. This could be due to budget-conscious single-item orders. Offering a complimentary two-piece Chicken 65 sample for first-time buyers could enhance satisfaction and encourage repeat business. Additionally, **Plain Biryani** showed mild positive correlation with popular add-ons like *Chicken Lollipop* and *Blueberry Goli Soda*, suggesting an opportunity to promote these through mapped add-on suggestions to increase basket size.

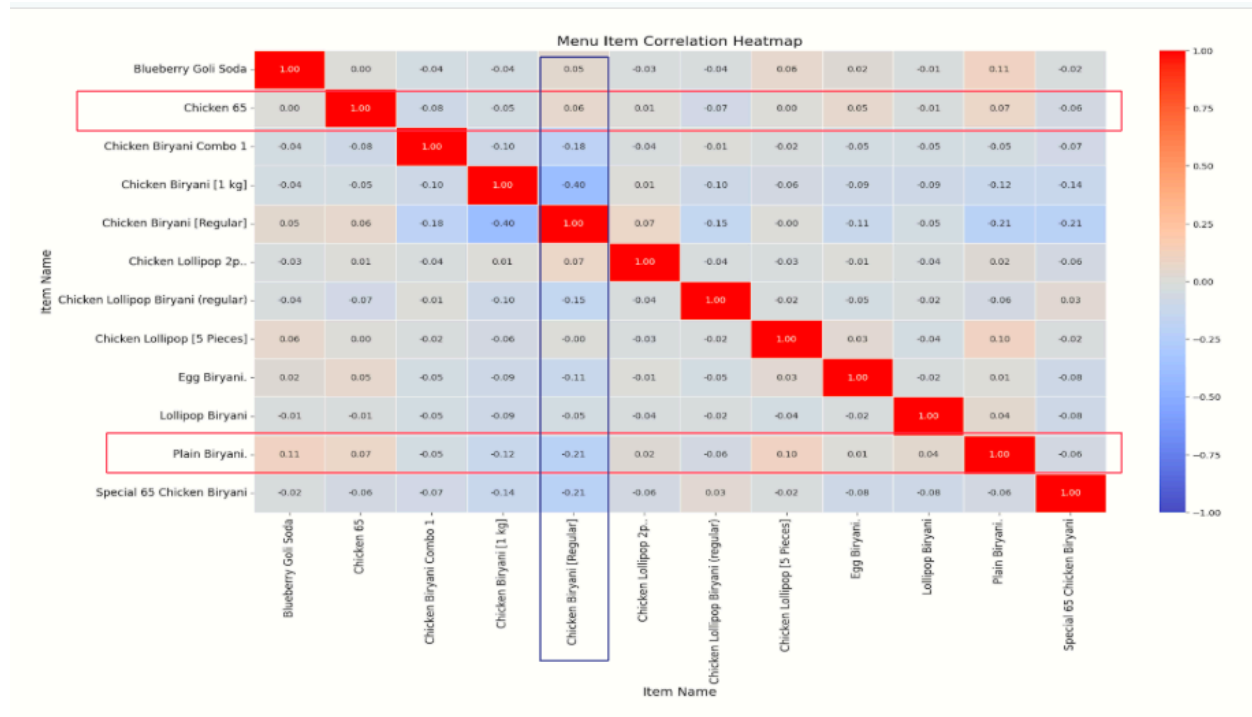


Figure 5 - Correlation Heatmap [Menu Items]

6.3 Customer Behavior Analysis

Customer retention emerged as a key concern, with a high churn rate of 96.43% and very few customers making repeat purchases across multiple months. On the other hand, the average Customer Lifetime Value (CLV) currently stands at ₹317.20, highlighting a significant opportunity for growth through better retention strategies.

These findings collectively provide a strong foundation for strategic improvements. They highlight trends in customer behavior, item popularity, and menu structure. Key insights point toward the need for simplifying the menu, improving communication around portions, refining combo offerings, and enhancing retention through personalized engagement.

In the final report, these insights will be further expanded with more in-depth analysis, predictive modeling, and targeted recommendations aimed at improving sales, brand recall, and addressing issues like delivery staff unavailability during peak hours.